Subject Description Form

Subject Code	DSAI5105
Subject Title	Hands-on AI for Science and Technology
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The objective of this course is to equip students with practical skills and algorithmic foundations in applying Artificial Intelligence (AI) to solve real-world problems in science and technology domains. The curriculum emphasizes hands-on implementation of AI techniques, covering machine learning, deep learning, and computational tools, with a focus on interdisciplinary applications in fields like science, engineering, healthcare, robotics, and others.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Identify key challenges and opportunities in applying AI to scientific research and technological innovation. b) Understand the usage of AI algorithms and tools and their strengths and weaknesses in different scientific problems. c) Demonstrate proficiency in implementing AI algorithms and tools for data analysis, modeling, and simulation. d) Analyze and derive insights from real-world case studies in science and technology using AI methods.
Subject Synopsis/ Indicative Syllabus	a) Fundamentals of AI for science: machine learning frameworks (PyTorch, TensorFlow), scientific computing (NumPy, SciPy). b) AI hands-on techniques for scientific data: regression, classification, clustering, image/signal processing, decision making and usage of LLMs. c) Case studies: Building large language model from scratch d) Challenges (data quality, model interpretability, computational efficiency) and emerging trends (AI-driven scientific discovery, federated learning for science, and ethical AI in tech.)

<table>
<tr><td>Teaching/Learning Methodology</td><td><u>Lectures:</u>
Introduce AI theories, tools, and interdisciplinary applications, with a focus on technical frameworks and scientific problem-solving.

<u>Tutorials:</u>
Hands-on coding sessions and group projects using real scientific datasets, simulation environments, and AI development tools.</td></tr>
<tr><td>Assessment Methods in Alignment with Intended Learning Outcomes</td><td>
<table>
<tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr>
<tr><td>a</td><td>b</td><td>c</td><td>d</td></tr>
<tr><td>1. Project</td><td>50 %</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>2. Exam</td><td>50 %</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>Total</td><td>100 %</td><td colspan="4"></td></tr>
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Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:
<u>Project:</u>
Students will complete an AI project addressing a scientific or technological problem (e.g., developing a model for protein structure prediction or autonomous robot navigation), demonstrating skills in data processing, algorithm implementation, and result analysis.
<u>Exam</u>
Written assessment covering AI theories, tool comparisons, and scenario-based problem-solving, evaluating understanding of AI’s role in science and technology.</td></tr>
<tr><td>Student Study Effort Expected</td><td>
<table>
<tr><td>Class contact:</td><td></td></tr>
<tr><td>▪ Lecture</td><td>26 Hrs.</td></tr>
<tr><td>▪ Tutorial</td><td>13 Hrs.</td></tr>
<tr><td>Other student study effort:</td><td></td></tr>
<tr><td>▪ Self-learning from reference materials</td><td>35 Hrs.</td></tr>
<tr><td>▪ Reading and research for the case study</td><td>40 Hrs.</td></tr>
<tr><td>Total student study effort</td><td>114 Hrs.</td></tr>
</table></td></tr>
<tr><td>Reading List and References</td><td>[1] Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
[2] Szepesvári, C. (2022). Algorithms for reinforcement learning. Springer nature.
[3] Shaheen, M. Y. (2021). Applications of Artificial Intelligence (AI) in healthcare: A review. Science Open Preprints.</td></tr>
</table>

	[4] Smith, P. D. (2018). *Hands-On Artificial Intelligence for Beginners: An introduction to AI concepts, algorithms, and their implementation*. Packt Publishing Ltd. [5] Wang, H., Fu, T., Du, Y., Gao, W., Huang, K., Liu, Z., ... & Zitnik, M. (2023). Scientific discovery in the age of artificial intelligence. *Nature*, *620*(7972), 47-60.